

COMP3334 - Tutorial

Cryptography

Outlines

This file is a guide to the installation of Python and its library, Jupyter.

The tutorial is based on Windows 11 (64-bit). The operations on different operating systems are similar.

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Python

Introduction

Python is a popular programming language, which is easy to learn and use.

Python has various kinds of libraries which can undertake different types of tasks. It is highly popular among programming languages. It ranks No.1 on the TIOBE index in January 2024, which measures the popularity of programming languages.

Jan 2024	Jan 2023	Change	Programming Language		Ratings	Change
1	1			Python	13.97%	-2.39%
2	2			C	11.44%	-4.81%
3	3			C++	9.96%	-2.95%
4	4			Java	7.87%	-4.34%
5	5			C#	7.16%	+1.43%

Python Installation

If you have already installed Python, please skip this section, and move to the next one. In this section, we will use Python 3.10.6 as an example.

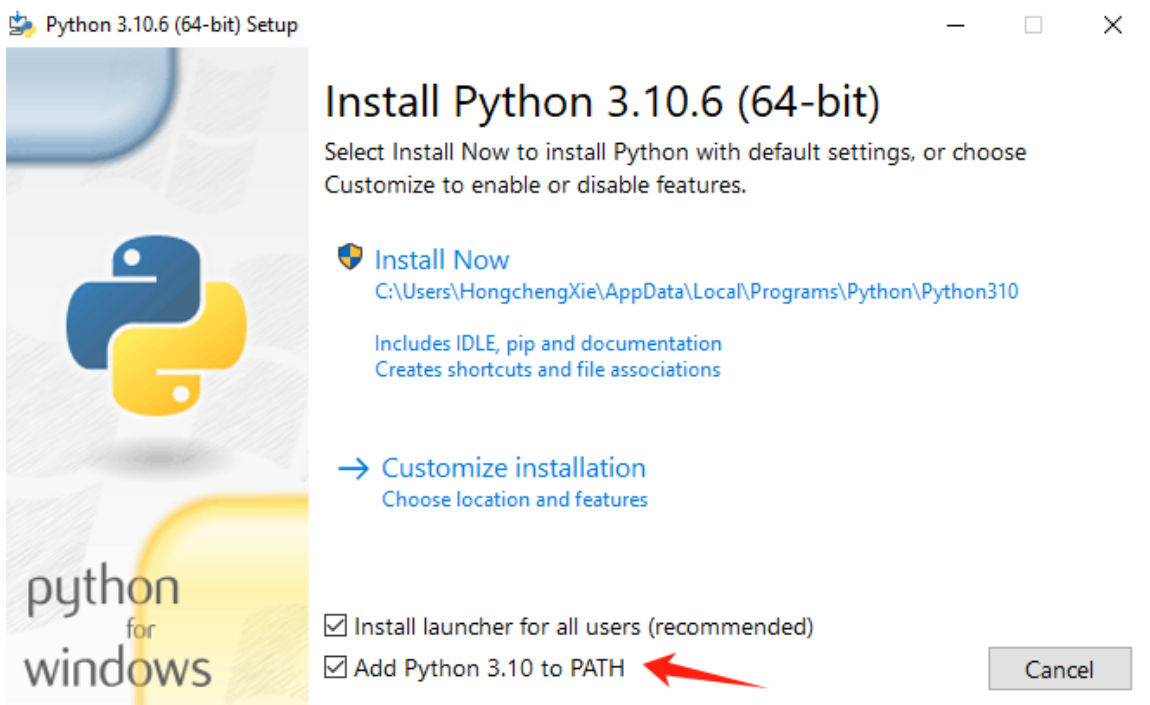
Note: In the remaining parts, the contents which are quoted from the computer and the commands we will input are in *Italics*.

1. Open your web browser and visit <https://www.python.org/downloads/release/python-3106/>.
2. Scroll down and find the installer which is for your operating system. For Windows 11 (64-bit), click *Windows installer (64-bit)* to download the installer.

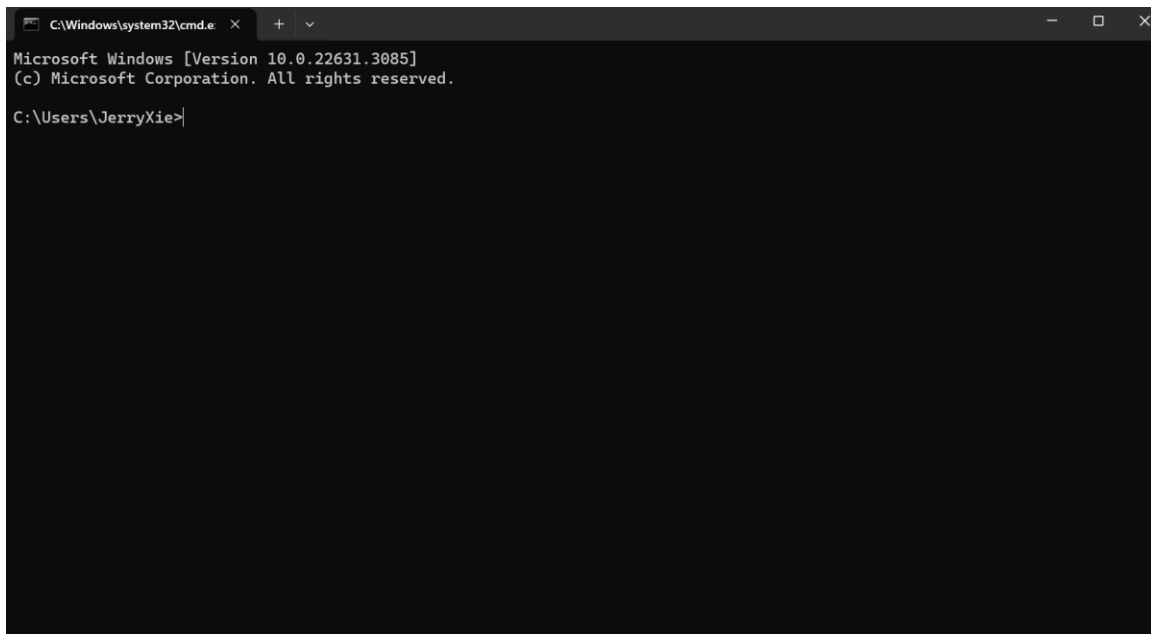
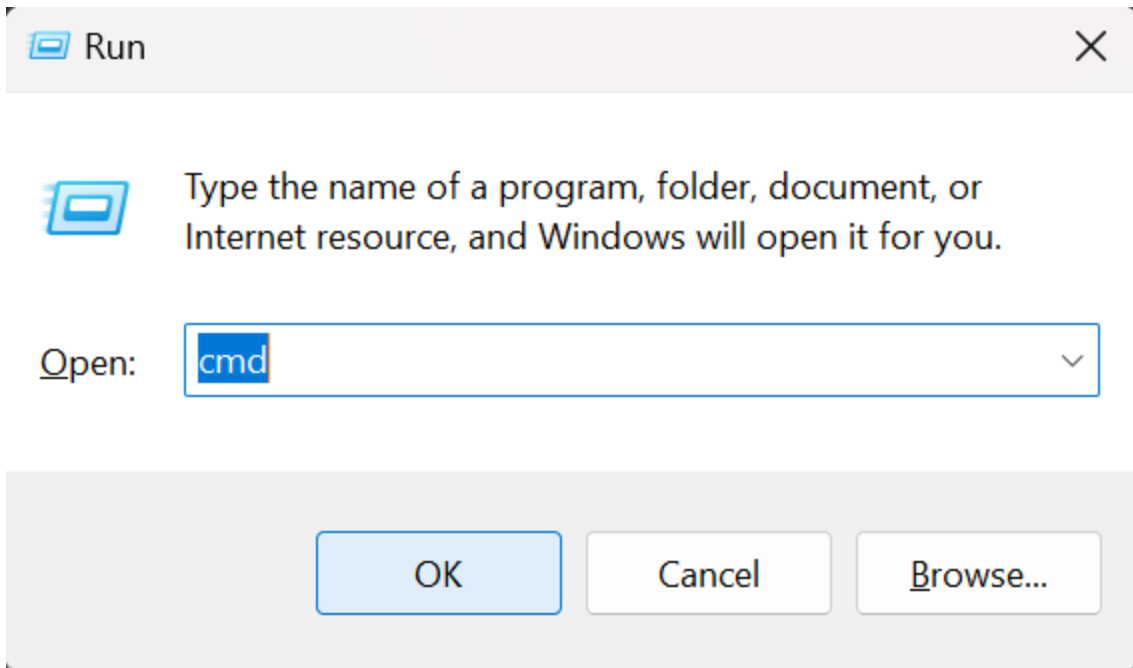
Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		d76638ca8bf57e44ef0841d2cde557a0	25986768	SIG
XZ compressed source tarball	Source release		afc7e14f7118d10d1ba95ae8e2134bf0	19600672	SIG
macOS 64-bit universal2 installer	macOS	for macOS 10.9 and later	2ce68dc6cb870ed3beea8a20b0de71fc	40826114	SIG
Windows embeddable package (32-bit)	Windows		a62cca7ea561a037e54b4c0d120c2b0a	7608928	SIG
Windows embeddable package (64-bit)	Windows		37303f03e19563fa87722d9df11d0fa0	8585728	SIG
Windows help file	Windows		0aee63c8fb87dc71bf2bcc1f62231389	9329034	SIG
Windows installer (32-bit)	Windows		c4aa2cd7d62304c804e45a51696f2a88	27750096	SIG
Windows installer (64-bit)	Windows	Recommended	8f46453e68ef38e5544a76d84df3994c	28916488	SIG

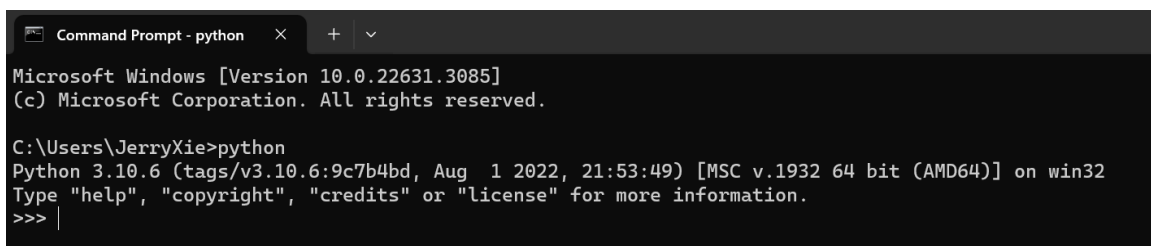
3. Open the installer you downloaded, tick *Add Python 3.10 to PATH*, and Click *Install Now* to install.



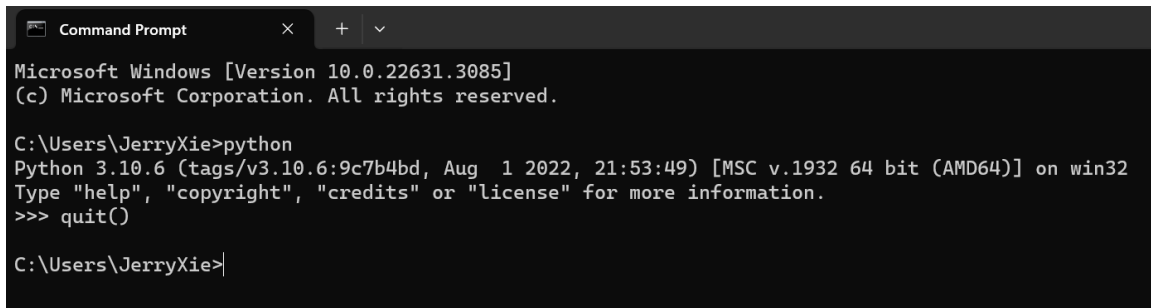
4. Press Win+R on your keyboard, type *cmd* and click *OK*, to open Command Prompt.



5. Type *python* and press Enter on your keyboard. Your installation will be successful if your python program is running.



6. You can type *quit()* and press Enter to exit Python program.

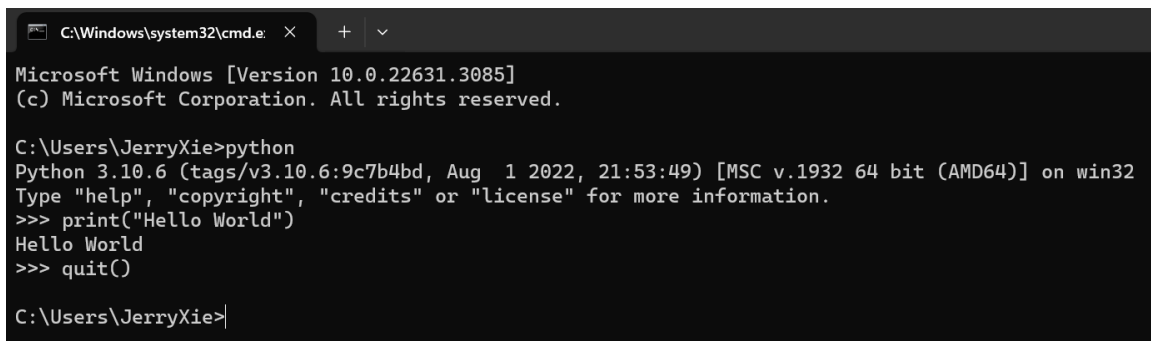


```
Command Prompt
Microsoft Windows [Version 10.0.22631.3085]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JerryXie>python
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> quit()

C:\Users\JerryXie>
```

Unlike C/C++ programs that are compiled into binary code, Python programs must be executed by Python's interpreter.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22631.3085]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JerryXie>python
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello World")
Hello World
>>> quit()

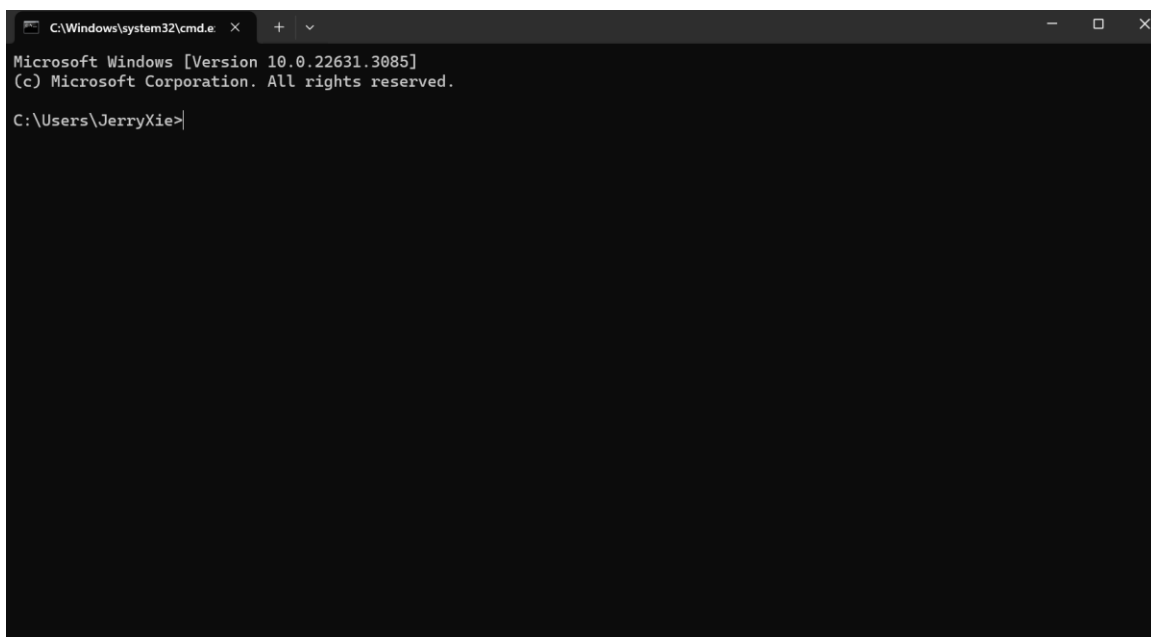
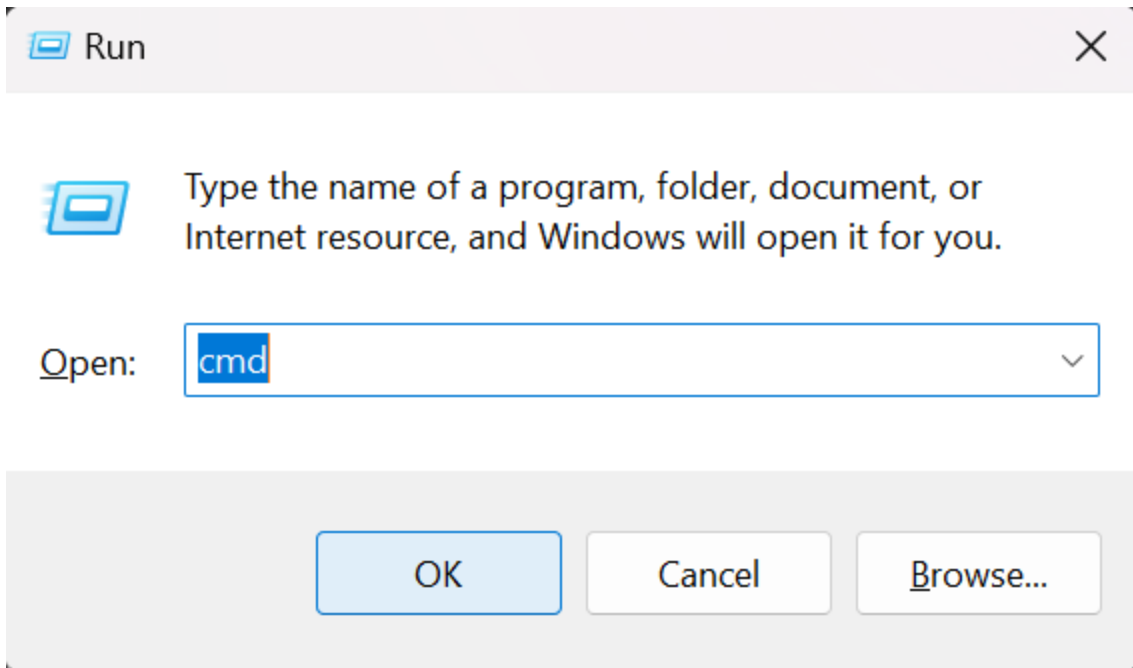
C:\Users\JerryXie>
```

Remarks: Python can also be installed in other ways, such as Anaconda. You can visit <https://www.anaconda.com/> for more information.

Library Installation

Python uses a package management system, pip, to manage the Python libraries in your computer. By using pip, you do not have to download libraries from the Internet manually. It can help you address the issues automatically.

1. Press Win+R on your keyboard, type *cmd* and click *OK*, to open Command Prompt.



2. To install a library, please type *pip install <library's name>* and press Enter.
(Please use the name of the library you want to install to replace the brackets).
3. To uninstall a library, please type *pip uninstall <library's name>* and press Enter.

We will use pip to install the libraries in the following sections.

Jupyter – A visualized Python notebook

Jupyter is a web platform which can be used to write code snippets, add documentation, and test programs all within a webpage.

Install Jupyter

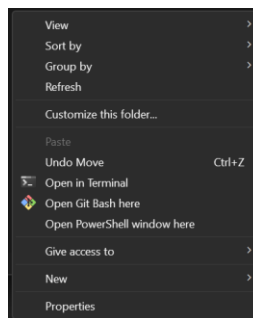
Jupyter can be installed by pip.

1. Open Command Prompt
2. Type `pip install jupyter` and press Enter.

```
C:\Users\JerryXie>pip install jupyter
Collecting jupyter
  Using cached jupyter-1.0.0-py2.py3-none-any.whl (2.7 kB)
Requirement already satisfied: ipywidgets in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from jupyter) (8.1.1)
Requirement already satisfied: jupyter-console in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from jupyter) (6.6.3)
Requirement already satisfied: qtconsole in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from jupyter) (5.5.1)
Requirement already satisfied: notebook in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from jupyter) (7.0.7)
Requirement already satisfied: nbconvert in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from jupyter) (7.14.2)
Requirement already satisfied: ipykernel in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from jupyter) (6.29.0)
Requirement already satisfied: matplotlib-inline>=0.1 in c:\users\jerryxie\appdata\local\programs\python\python310\lib\site-packages (from ipykernel->jupyter) (0.1.6)
```

Launch Jupyter Notebook

1. Open the tutorial folder, hold down Shift on your keyboard and right-click in a blank space. Choose *Open in Terminal* to open Command Prompt in this folder.



2. Type `jupyter notebook` and press Enter to launch Jupyter.

```
(venv) C:\Users\JerryXie\Desktop\jupyter>jupyter notebook
```

3. You can find the notebooks under *Files* tab.
4. Click a notebook with the suffix `.ipynb` to open the tutorial notebook. If a window with the title *Select Kernel* appears, click *Select* button.

A notebook consists of some cells.

- Markdown cells can be used to write documentation and texts by using Markdown language.
 - Markdown: Please refer to <https://www.markdownguide.org/basic-syntax/>
- Python cells can be used to write code in Python. The running result will be shown under this cell.

Running on a Markdown cell will let this cell show the content in the specific format defined in Markdown.

Tutorial

Please follow the instructions in the Jupyter notebook of our tutorial.